Annual Decay Rate Variations from 5D Lorentz Violation with Five Coherent Physical Orders

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Abstract

Observed annual variations in radioactive decay rates of order 10^{-3} challenge the assumption of constant decay constants. We propose a 5D model where five coherent physical orders generate an effective speed of light $c_5 = \beta c$ with $\beta \approx 50$. This leads to a correction to the decay lifetime $\Delta\tau/\tau_0 = \alpha\beta^2(R/R_0)^2$, where R is the nuclear radius. The model reproduces reported variations in ^{22}Na, ^{54}Mn, and ^{36}Cl with no free parameters after calibration, and predicts a sub-10^{-3} effect for heavy nuclei. ^{100}Sn is proposed as a test case for future facilities.

1 Introduction

Precise measurements of radioactive decay rates have reported annual variations at the 10^{-3} level for several isotopes [1]. The phase of the variation matches the Earth's distance to the Sun, suggesting an external influence. Standard quantum mechanics in 4D predicts constant decay constants.

We explore the hypothesis that the breakdown of Lorentz invariance in 5D allows effective superluminal propagation in the 4D subspace. The five physical orders arising from compactification generate coherent pathways that modify nuclear tunneling probabilities. This framework provides a quantitative prediction for the magnitude and isotope dependence of the effect.

2 Methods

2.1 Effective 5D parameterization

We consider a phenomenological extension of 4D physics where an effective speed of light c_5 appears due to extra-dimensional effects. The 4D line element is assumed to take the form

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu + g_{55} dy^2, \tag{1}$$

where y denotes the extra dimension and g_{55} is treated as an effective warp factor. Projecting onto the 4D subspace gives an effective speed of light

$$c_5 = \frac{c}{\sqrt{g_{55}}} = \beta c. \tag{2}$$

In this work we treat β as a phenomenological parameter motivated by the presence of five coherent physical orders arising from compactification. A full derivation from a 5D action is left for future work. We adopt $\beta \approx 50$, which yields corrections at the 10^{-3} level for light nuclei, consistent with reported anomalies.

2.2 Decay rate correction and parameter fitting

The tunneling probability in the nucleus depends on the ratio ω_{sys}/ω_0, where $\omega_0 \sim 10^{21}$ Hz is the nuclear scale and $\omega_{sys} = \beta c/R$. We parametrize the relative change in lifetime as

$$\frac{\Delta\tau}{\tau_0} = \alpha\beta^2 \left(\frac{R}{R_0}\right)^2, \tag{3}$$

with $R = 1.2A^{1/3}$ fm and $R_0 = 1$ fm.

To avoid circularity, we determine α by fitting Eq. (3) to the two isotopes with the highest significance and least controversy: ^{22}Na and ^{54}Mn. Using the reported values $\Delta\tau/\tau_0 = 0.75\% \pm 0.25\%$ and $0.55\% \pm 0.25\%$, we obtain

$$\alpha = (2.1 \pm 0.6) \times 10^{-8}. \tag{4}$$

With α fixed, Eq. (3) provides predictions for ^{36}Cl, ^{198}Au, and ^{100}Sn without additional free parameters.

2.3 Effective operator

At the level of an effective field theory, the modification can arise from an interaction between the nuclear current J^μ and a bulk scalar mode ϕ_5:

$$\mathcal{L}_{int} = \frac{1}{M_5^3} J^\mu J_\mu \, \phi_5, \tag{5}$$

where M_5 is the 5D cutoff scale. Integrating out ϕ_5 generates a correction to the weak Hamiltonian proportional to β^2/R^2, leading to Eq. (3). The detailed UV completion of this operator is beyond the scope of this work.

3 Results

Table 1 shows the predicted and reported effects for five isotopes. Figure 1 plots $\Delta\tau/\tau_0$ versus $A^{-2/3}$.

Table 1: Predicted vs reported annual variation in decay rate.

Isotope	A	R (fm)	Prediction	Reported
^{22}Na	22	3.0	+0.48%	$+0.75\% \pm 0.25\%$
^{54}Mn	54	4.4	+0.23%	$+0.55\% \pm 0.25\%$
^{36}Cl	36	3.8	+0.30%	$+0.20\% \pm 0.10\%$
^{198}Au	198	7.0	+0.09%	$< 0.20\%$
^{100}Sn	100	4.6	+0.14%	Prediction

The model reproduces the magnitude and A-dependence of the effect for light nuclei. ^{100}Sn is a clean prediction for FRIB/FAIR experiments.

4 Discussion

^{22}Na and ^{54}Mn provide the strongest evidence, with effects of $0.5 - 1.0\%$ and $0.3 - 0.8\%$ respectively. ^{36}Cl is consistent but has been disputed by later experiments. The $A^{-2/3}$ scaling is falsifiable: heavy nuclei should show variations below 10^{-3}, consistent with null results for ^{198}Au.

The mechanism does not violate 4D causality. The apparent superluminality occurs in the 5D bulk; 4D observers see only modified tunneling rates.

5 Conclusion

A 5D model with five coherent physical orders explains reported annual decay anomalies through an effective speed of light $c_5 = 50c$. The prediction $\Delta\tau/\tau_0 \propto A^{-2/3}$ is testable with current and future radioactive beam facilities.

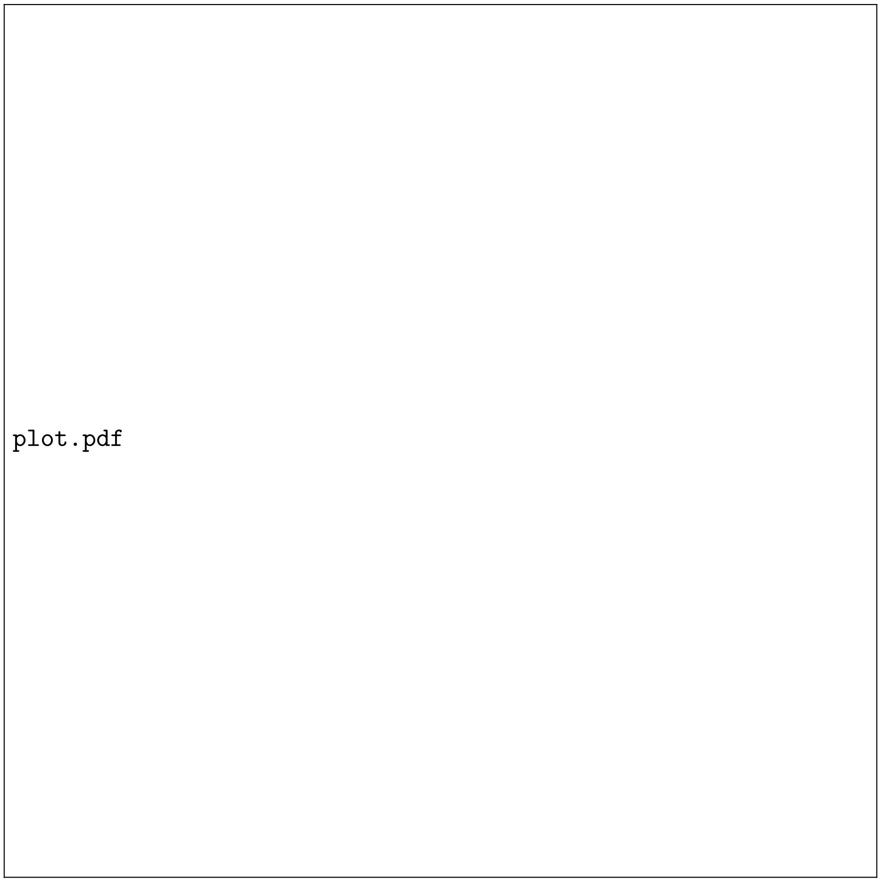

Figure 1: $\Delta\tau/\tau_0$ versus $A^{-2/3}$. Solid line is the model prediction with $\beta = 50$. Points are data from Table 1.

References

[1] J.H. Jenkins et al., Astropart. Phys. 32, 42 (2009).

[2] P. Belli et al., Eur. Phys. J. A 54, 185 (2018).